



Data Science: Exploratory Data Analysis of Airline Delays and Cancellations

Instructor: Dr. Kheradpisheh
Shahid Beheshti University — Spring 2026

1 Introduction

This project asks you to carry out a structured exploratory data analysis (EDA) on United States domestic flights between 2009 and 2018, using the Kaggle dataset *Airline Delay and Cancellation Data, 2009–2018* by Yuanyu Wendy Mu.¹ The data describe individual flights with information on carriers, airports, times, delays, cancellations, and delay causes. Working with such records is a common task in applied statistics and data science, since operational systems often store detailed logs over long periods.

In this project you will describe the structure of the dataset, identify data quality issues, construct and interpret graphical summaries, and explain your analytical choices in a clear written report supported by figures and tables.

2 Dataset and tools

2.1 Dataset

You must use the Kaggle dataset *Airline Delay and Cancellation Data, 2009–2018*.² The main data are provided as one CSV file per year from 2009 to 2018. Key variables include flight date, operating carrier, flight number, origin and destination airport, scheduled and actual departure and arrival times, departure and arrival delay in minutes, cancellation and diversion indicators, elapsed times, distance, and several fields that break down delay causes.

You may consult external descriptions of the dataset for more detail, such as the documentation in the repository *Kaggle-Airline-Data-Analysis*.³ Do not copy text directly; instead, summarise in your own words.

2.2 Computing environment

You are encouraged to use Kaggle Notebooks as your primary environment, since they provide direct access to the dataset and a ready-to-use Python setup.⁴ You may also work locally if you prefer, as long as your notebook runs to completion and reproduces your figures.

Your code will rely mainly on:

- pandas for tabular data manipulation.
- A plotting library such as matplotlib, seaborn, or plotly.

3 Project tasks: exploratory data analysis

This section outlines the core EDA components that your notebook and report should cover. You can add more, but these elements are expected for all projects.

3.1 Data understanding and preparation

Start by describing the structure of the data. In your notebook, show outputs of `head()`, `info()`, and `describe()` on a representative sample. In your report, summarise:

- The number of rows and columns you work with.
- The main variables, grouped into categories such as identifiers, time variables, delay measures, and indicators.
- Any transformations you apply, such as converting time fields or casting carrier codes to categorical types.

Include a small table listing the key fields you analyse, with brief descriptions.

¹<https://www.kaggle.com/datasets/yuanyuwendymu/airline-delay-and-cancellation-data-2009-2018>

²Dataset page and files: <https://www.kaggle.com/datasets/yuanyuwendymu/airline-delay-and-cancellation-data-2009-2018>

³<https://github.com/aroraayush/Kaggle-Airline-Data-Analysis>

⁴Public notebooks for this dataset: <https://www.kaggle.com/yuanyuwendymu/airline-delay-and-cancellation-data-2009-2018/code>

3.1.1 Missing data and data quality

Compute the proportion of missing values for the main variables, in counts and percentages. Discuss the pattern of missingness. For example, comment on when delay cause fields are populated and when they are zero or missing, and how you interpret those cases.

Explain how you handle missing values in different parts of the EDA. For instance, you might remove rows with missing arrival delay when studying delay distributions but keep them when counting flights by carrier. Make these decisions explicit.

3.1.2 Derived variables

Define a small set of derived variables that will support your analysis. Examples:

- A binary indicator for flights with arrival delay at least 15 minutes.
- A categorical time-of-day variable based on scheduled departure time.
- A measure of total delay minutes or a field describing the main delay cause for each delayed flight.

Describe these variables in the notebook (as code with short comments) and in the report (in a few sentences). Use them consistently in later tables and plots. keep in mind these are suggestions, feel free to experiment as long as you can explain the thought process behind it.

3.2 Univariate analysis

3.2.1 Flight counts and distributions

Produce descriptive statistics and plots for key variables. At minimum:

- A table or bar chart showing number of flights per year and per month.
- A histogram or density plot of arrival delay, and a similar plot for departure delay.
- A histogram or density plot of distance.

For each plot, write a short interpretation. Comment on the shape of the distribution, presence of outliers, and any notable features.

3.2.2 Categorical variables

Summarise categorical variables such as carriers and airports. Suggested components:

- A bar chart of the top ten carriers by number of flights.
- A bar chart of the top origin airports by departures, or destination airports by arrivals.
- A table reporting the proportion of cancelled and diverted flights.

Relate these summaries to the context of domestic air travel, for example by noting which carriers or airports dominate the data.

3.3 Time patterns

Investigate how delays and cancellations change over time. Construct plots that show:

- Average arrival delay by month across the period you analyse.
- Number of flights per month.
- Average arrival delay by year.

Also examine cancellation rates over time by computing the fraction of flights cancelled per month or year. Present these in line charts and discuss patterns that you observe.

3.3.1 Day of week and time of day

Set up views of delay by day of week and time of day. For example:

- A bar chart of average arrival delay by day of week.
- A plot of average delay by hour of scheduled departure or by time-of-day category.

Comment on which days or time periods show higher delays and offer possible explanations, while keeping in mind that you are not performing causal analysis.

3.4 Carriers, airports, and routes

Compare delay and cancellation behaviour between airlines. At least:

- Compute and plot average arrival delay by carrier.
- Compute and plot cancellation rates by carrier.

Rank carriers by these metrics and discuss which appear to perform better or worse. You can look at examples in public repositories for ideas on how to structure these summaries.⁵

3.4.1 Origin and destination patterns

Study how delays vary across airports. You can:

- Identify the busiest origin airports and compute their average delays.
- Do the same for destination airports.
- Optionally, look at a few busy routes (origin–destination pairs) and compare their delay statistics.

You may keep this at the level of tables and bar charts. If you wish to add airport names and locations, you can merge with an external airport metadata table such as the one in the Kaggle-Airline-Data-Analysis repository⁶, also you can experiment with some better visualizations such as a heat map over the locations.

3.5 Relationships between variables

Produce at least two plots that explore relationships between numeric variables, for example:

- A scatter or hexbin plot of departure delay versus arrival delay.
- A plot of distance versus arrival delay, using sampling or binning if needed.

Discuss whether the relationships seem strong or weak, whether there are clear patterns, and whether certain ranges behave differently from others.

3.5.1 Correlation and causes

Compute a correlation matrix for selected numeric variables, such as the various delay components. Display this matrix as a heatmap and comment on which components show stronger associations.

Also include at least one summary of delay causes. For example, compute the total minutes of delay attributed to each cause across your subset, or the proportion of delayed flights where a given cause is present. Present this as a table or bar chart and interpret what it suggests about the main sources of delay.

4 Project outputs

You will submit two files: a written report in PDF and a Jupyter notebook.

⁵For instance, https://github.com/Hyshubham2504/Airline_Delay_and_Cancellation_Analysis

⁶<https://github.com/aroraayush/Kaggle-Airline-Data-Analysis>

4.1 Structure

The report should be six to ten pages long (excluding references and any appendices) and use the structure below:

1. **Introduction.** Describe the dataset, the context of airline delays, and the aim of your analysis.
2. **Data and methods.** Explain the data source, the variables you focus on, your preprocessing steps, and your general approach to summarising and visualising the data. Include a table of key variables and describe any derived variables.
3. **Exploratory analysis.** Present your main findings, organised into subsections such as data overview, time patterns, carrier and airport comparisons, and relationships between variables. Integrate figures and tables into the discussion.
4. **Discussion and limitations.** Reflect on what your EDA reveals about airline performance and on any limitations of the dataset or your approach.
5. **Bonus work.** If you carried out optional extensions, describe them briefly here.
6. **Conclusion.** Summarise the main insights and possible next steps.
7. **References.** List the dataset and any external tutorials, notebooks, or articles you relied on.

4.2 Jupyter notebook

The notebook should:

- Start with a markdown cell giving the project title, your name, the dataset source, and a brief description of the task.
- Use markdown headings that roughly match the report sections.
- Contain all code needed to reproduce the tables and figures used in the report.
- Be written so that it can be run from top to bottom without manual intervention.

Place imports at the top. Use comments where code might not be clear at first glance. If you publish a version of your notebook on Kaggle, include the link in your report.

5 Bonus work (optional)

The following extensions are optional. They can be mentioned in the report under the bonus section and may contribute extra credit.

- You may build a basic predictive model that uses pre-departure information to predict whether a flight will be delayed beyond a threshold such as 15 minutes.
- Comparing the distribution of causes across carriers, airports, or seasons.
- Studying how the composition of delay causes changes over the years.
- Looking at flights with multiple delay causes and summarising their characteristics.
- Merge with an airport metadata table that adds airport names, cities, and coordinates, such as the one used in the Kaggle-Airline-Data-Analysis repository.⁷
- If time allows, bring in simple weather summaries by region and date from a public source and link them to flight dates.

⁷<https://github.com/aroraayush/Kaggle-Airline-Data-Analysis>

- If you are interested in data engineering aspects, Use Dask to perform grouping and aggregation operations on the full ten year dataset, following patterns outlined in tutorials.⁸
- Convert the yearly CSV files to Parquet and compare loading and processing times.

When describing these extensions, focus on clear explanations and simple comparisons, make sure you have a **bonus** section in your report and mention anything that you have done and think deserves more attention.

Stay hungry, stay foolish

⁸Dask example: <https://tilburgsciencehub.com/topics/manage-manipulate/loading/large-datasets/dask-in-action/>